

B. Create a Student Record Management PWA.

CODE:

(Index.cshtml):

```
@{
    ViewData["Title"] = "Home Page";
}

<div class="text-center">
    <h1 class="display-4">Welcome</h1>
    <p>Learn about <a href="https://learn.microsoft.com/aspnet/core">building Web
apps with ASP.NET Core</a>.</p>
    <h1>Offline Page</h1>
    <h2>Student Record</h2>

    <p>Roll No : @Model.RollNo</p>

    <p>Name : @Model.Name</p>
</div>
```

(HomeController.cs):

```
using Microsoft.AspNetCore.Mvc;
using Practical_7A.Models;
using System.Diagnostics;

namespace Practical_7A.Controllers
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            Student s = new Student();

            s.RollNo = "T016";
            s.Name = "Himani Malankar";

            return View(s);
        }

        public IActionResult Privacy()
        {
            return View();
        }

        [ResponseCache(Duration = 0, Location = ResponseCacheLocation.None, NoStore = true)]
        public IActionResult Error()
        {
            return View(new ErrorViewModel { RequestId = Activity.Current?.Id ?? HttpContext.TraceIdentifier });
        }
    }
}
```

(Student.cs):

```
using Microsoft.AspNetCore.Mvc;
using Practical_7A.Models;
using System.Diagnostics;
```

```

namespace Practical_7A.Controllers
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            Student s = new Student();

            s.RollNo = "T016";
            s.Name = "Himani Malankar";

            return View(s);
        }

        public IActionResult Privacy()
        {
            return View();
        }

        [ResponseCache(Duration = 0, Location = ResponseCacheLocation.None, NoStore
= true)]
        public IActionResult Error()
        {
            return View(new ErrorViewModel { RequestId = Activity.Current?.Id ??
HttpContext.TraceIdentifier });
        }
    }
}

```

(_Layout.cshtml):

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>@ViewData["Title"] - Practical_7A</title>
    <script type="importmap"></script>
    <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
    <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
    <link rel="stylesheet" href="~/Practical_7A.styles.css" asp-append-version="true" />
    <link rel="manifest" href="manifest.json"/>
</head>
<body>
    <header>
        <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light
bg-white border-bottom box-shadow mb-3">
            <div class="container-fluid">
                <a class="navbar-brand" asp-area="" asp-controller="Home"
asp-action="Index">Practical_7A</a>
                <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target=".navbar-collapse" aria-controls="navbarSupportedContent"
aria-expanded="false" aria-label="Toggle navigation">
                    <span class="navbar-toggler-icon"></span>
                </button>
                <div class="navbar-collapse collapse d-sm-inline-flex
justify-content-between">
                    <ul class="navbar-nav flex-grow-1">
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Index">Home</a>
                        </li>
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>

```

```

        </li>
      </ul>
    </div>
  </div>
</nav>
</header>
<div class="container">
  <main role="main" class="pb-3">
    @RenderBody()
  </main>
</div>

<footer class="border-top footer text-muted">
  <div class="container">
    &copy; 2026 - Practical 7A - <a asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
  </div>
</footer>
<script src="~/lib/jquery/dist/jquery.min.js"></script>
<script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
<script src="~/js/site.js" asp-append-version="true"></script>
@await RenderSectionAsync("Scripts", required: false)
<script>
  if('serviceWorker' in navigator)
  {
    navigator.serviceWorker.register('/service-worker.js')
  }
</script>
</body>
</html>

```

(service-worker.js):

```

const CACHE = "v1";

self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/Home/Index",
        "/css/site.css",
        "/js/site.js"
      ]);
    })
  );
});

self.addEventListener("fetch", e => {
  e.respondWith(
    caches.match(e.request).then(response => {
      return response || fetch(e.request);
    })
  );
});

```

(manifest.json):

```

{
  "name": "PWA Offline Demo",
  "short_name": "PWA",
  "start_url": "index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#2196f3",
  "icons": [

```

```
{
  "src": "/Icons/icon1.png",
  "sizes": "128x128",
  "type": "image/png",
  "purpose": "any"
},
{
  "src": "/Icons/icon2.png",
  "sizes": "512x512",
  "type": "image/png",
  "purpose": "any"
}
]
```

OUTPUT:

